Exp 8: Configuring User Management and Role-Based Access Control (RBAC) in a Zoho Creator Application.

Register Number: 241401127

1. Aim

To configure user management and role-based access control (RBAC) in the "Loan Application & Pre-Screening Management" application built in Zoho Creator, so that the Administrator, Loan Officer, and Applicant roles each have different levels of access to the Loan_Application_Form and its records.

2. Procedure

2.1 Problem Statement & Access Control Scope

In a bank, the people who use a loan application system have different responsibilities. Applicants should only submit and view their own applications. Loan officers need to review and process all applications submitted to their branch. The administrator needs full control over the application, including users, roles, and settings. Without access control, every user could see and modify every applicant's personal and financial data, which breaks data privacy requirements.

This experiment designs a three-level RBAC model for the application using Zoho Creator's built-in user management features.

- **Application:** Loan Application & Pre-Screening Management
- **Protected Form:** Loan_Application_Form
- **Protected Report:** Loan_Application_Form_Report

User Management Location: Settings → User Permissions (Permissions, Roles, Data Sharing tabs)

2.2 Access Control Model

The access model uses three roles arranged in a hierarchy, with each role linked to a permission profile and a data-sharing rule.

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Role	Purpose	Form Access	Report Access	Permitted Actions
CEO (Administrator)	Full control of the application	Create, View, Edit, Delete	All records	Manage users, roles, workflows, and settings
Loan_Officer	Reviews and processes loan applications	View and Edit	All applicant records (Read/Write via sharing rule)	Send document reminders, push leads to CRM
Applicant	Submits a loan enquiry	View and Create	Own records only	No administrative actions

2.3 Default Permission Profiles Review

On opening **Settings → User Permissions → Permissions**, Zoho Creator provides two default profiles:

- **Read:** Read permission for all components

Write: Write permission for all components

These defaults were reviewed before creating custom profiles.

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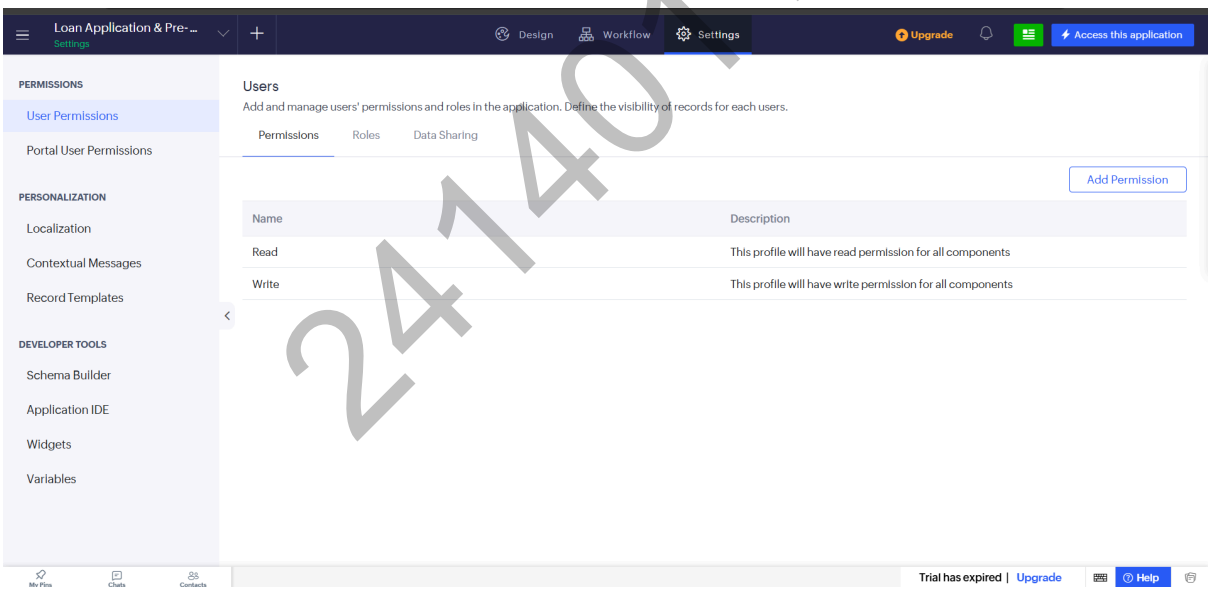


Figure 1: User Permissions screen showing the default Read and Write permission profiles

2.4 Role Hierarchy Configuration

The default role **CEO** sits at the top of the hierarchy. Two new roles were added beneath it:

1. **Loan_Officer**, reporting to CEO

Applicant, reporting to Loan_Officer

Resulting hierarchy:

In Zoho Creator, a role higher in the hierarchy can access the records of roles below it, so the hierarchy supports the intended supervision model.

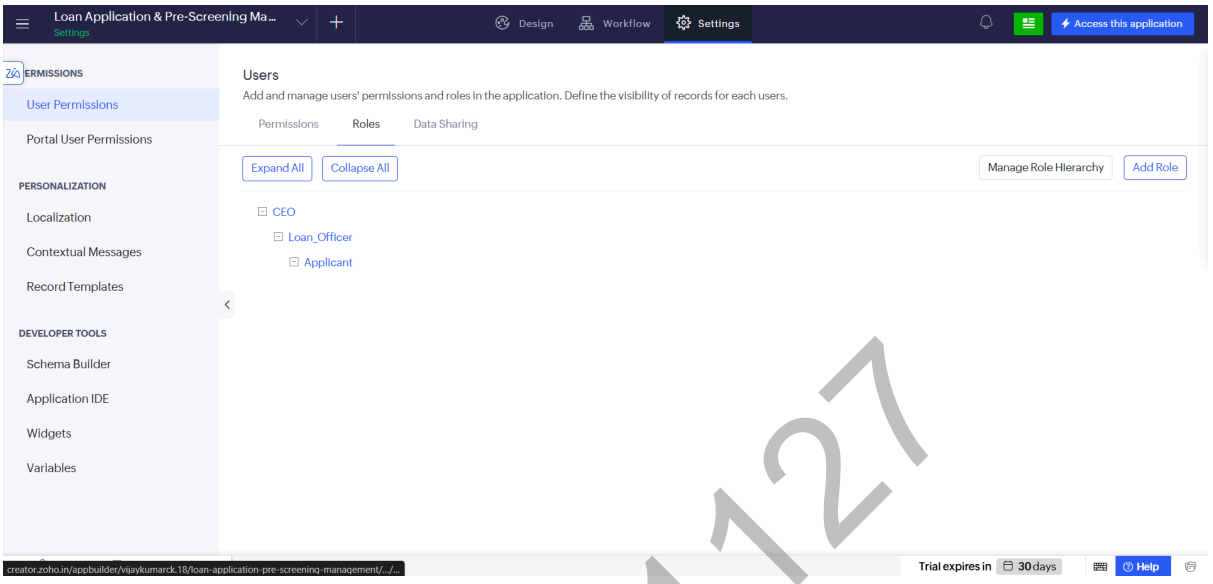


Figure 2: Roles tab showing the CEO → Loan_Officer → Applicant hierarchy

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2.5 Custom Permission Profiles

Two custom permission profiles were created to match the roles.

Loan_Application_Form: Access enabled, View and Edit enabled, Delete disabled, More disabled
- **Loan_Application_Form:** Access enabled, View and Edit enabled, Delete disabled, More disabled
Loan_Officer_Dashboard: Access enabled

Applicant_Permission (description: "Loan applicant")

- Scoped to Loan_Application_Form so the applicant can submit and view the form

Loan_Officer_Dashboard access disabled, so applicants cannot see the officer's dashboard

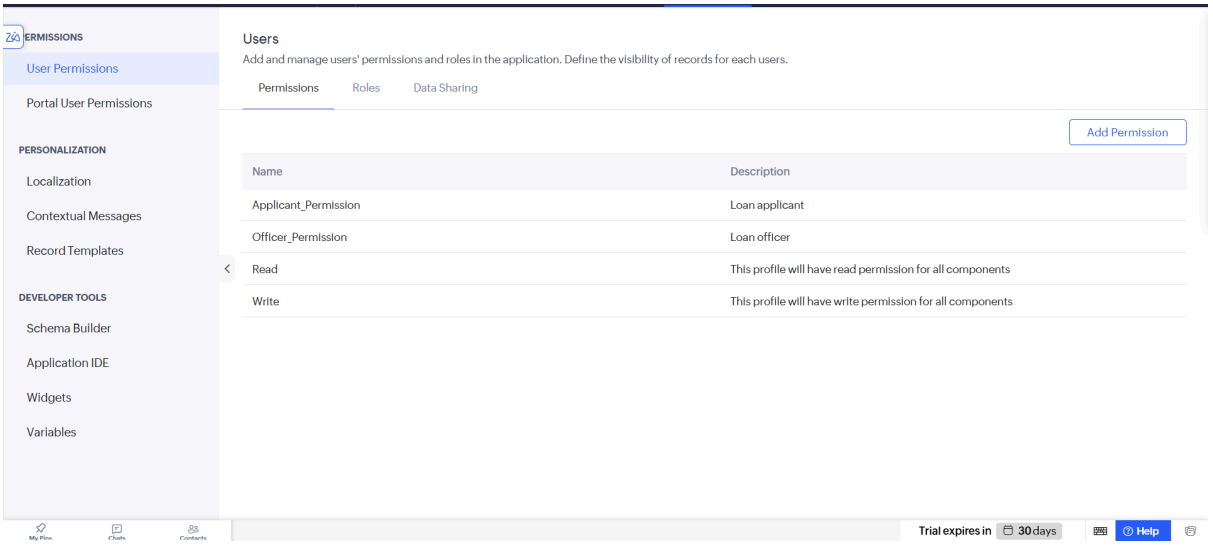


Figure 3: Permissions list showing Applicant_Permission, Officer_Permission, Read, and Write

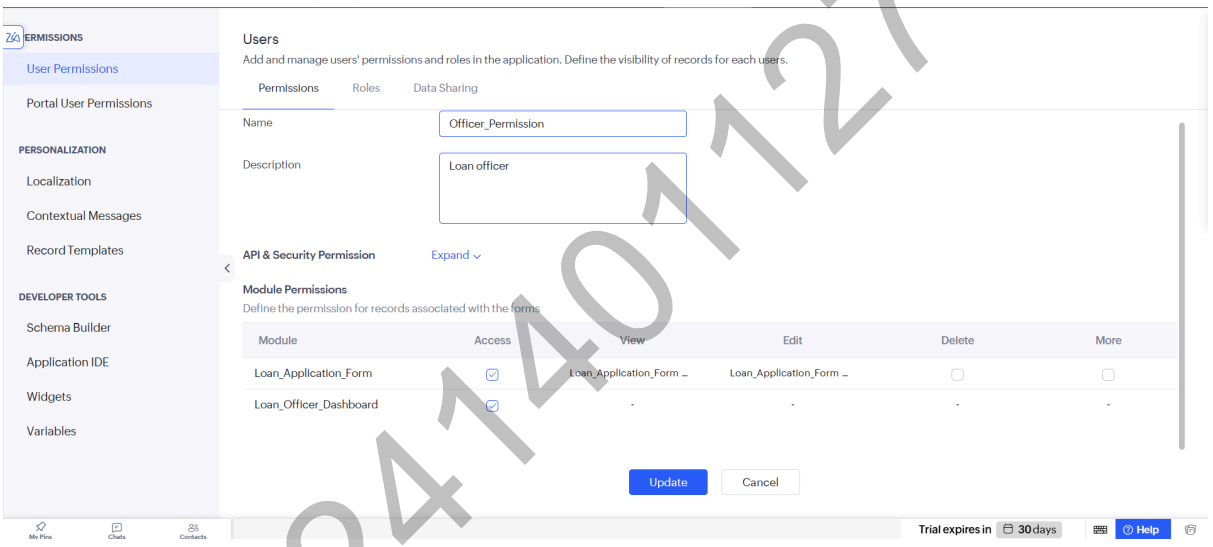


Figure 4: Officer_Permission module grid showing Access, View, Edit, and Delete settings for Loan_Application_Form and Loan_Officer_Dashboard

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2.6 Test User Assignment

Two test users were added through **Creator Home → Users → Add User**, each assigned a role and a permission profile:

Test User	Role	Permission Profile
Test User 1	Loan_Officer	Officer_Permission
Test User 2	Applicant	Applicant_Permission

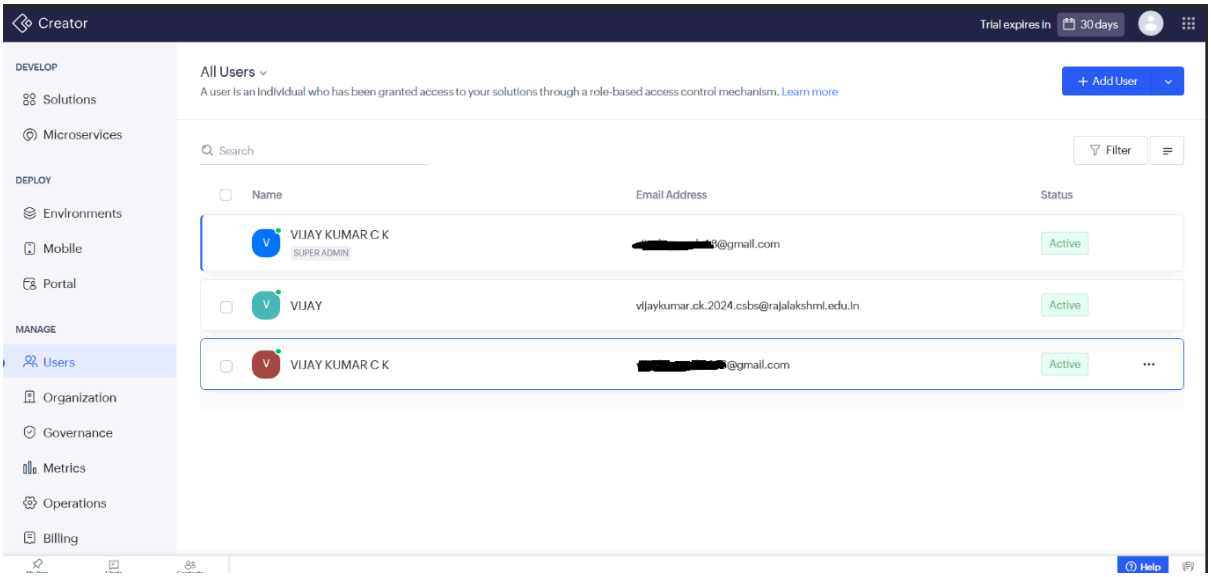


Figure 5: Users list showing the added test users with assigned roles

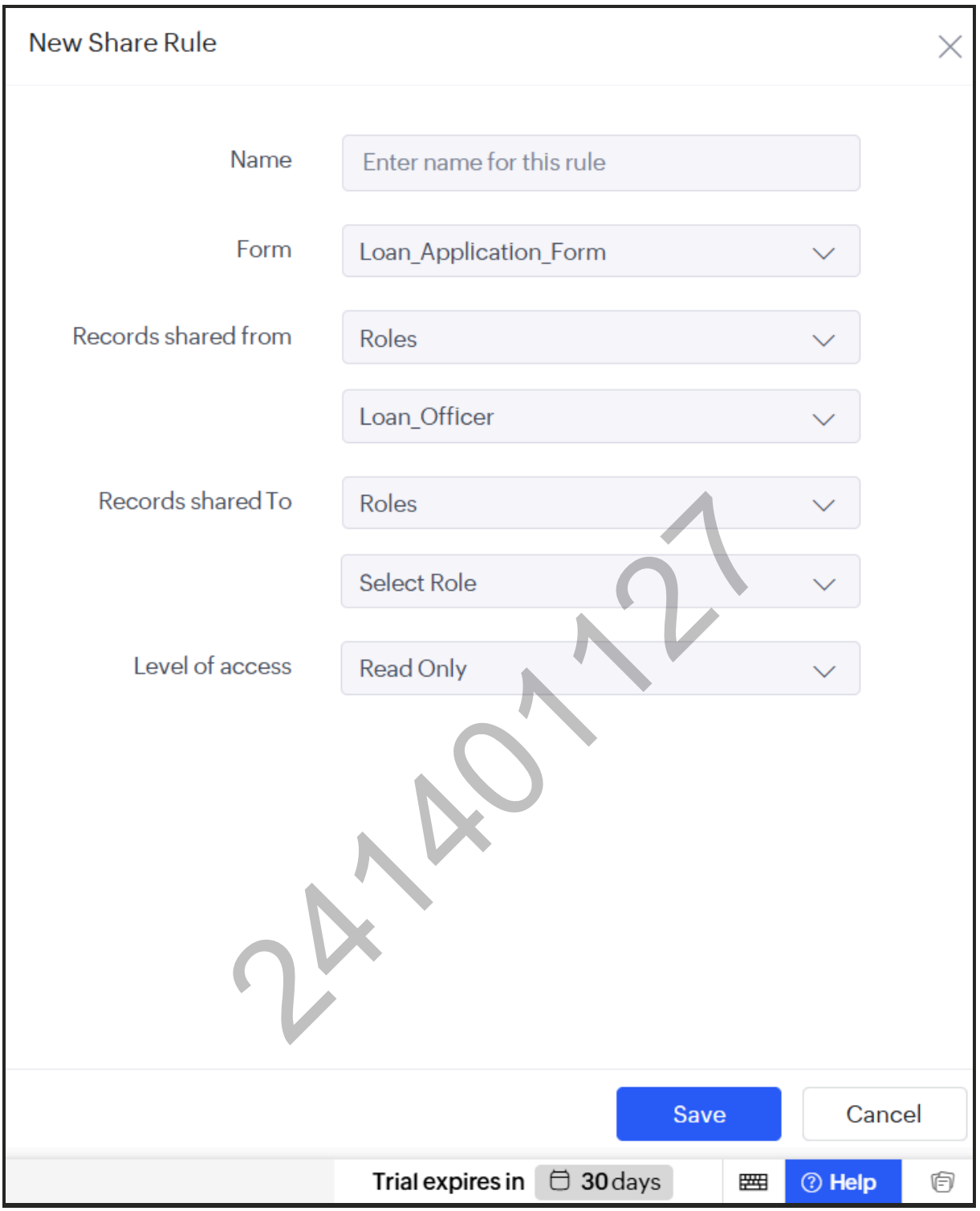
2.7 Data Sharing Rule Configuration

Role permissions control what a user can do. Data sharing rules control which records a user can see. A share rule was created so that records submitted by Applicants are visible to Loan Officers:

Setting	Value
Rule Name	Applicant_To_Officer_Share
Form	Loan_Application_Form
Records Shared From	Role: Applicant
Records Shared To	Role: Loan_Officer
Level of Access	Read/Write

Because no sharing rule exists for Applicant-to-Applicant visibility, each applicant sees only the records they created.

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Figure 6: New Share Rule form with the above configuration

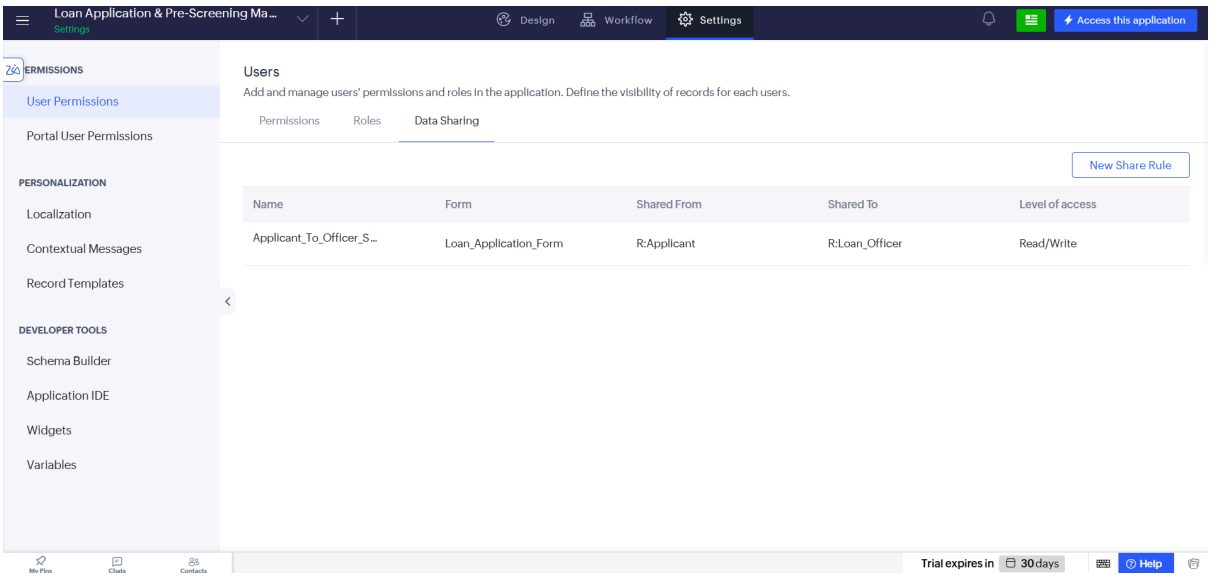


Figure 7: Data Sharing tab listing the saved rule (R:Applicant → R:Loan_Officer, Read/Write)

2.8 Live Access Test (Planned)

The intended verification steps are:

1. Log in as the Applicant test user and submit a new loan application.
2. Log in as the Loan_Officer test user and confirm the submitted record appears in the report.

Confirm the Applicant test user cannot see records submitted by other applicants.

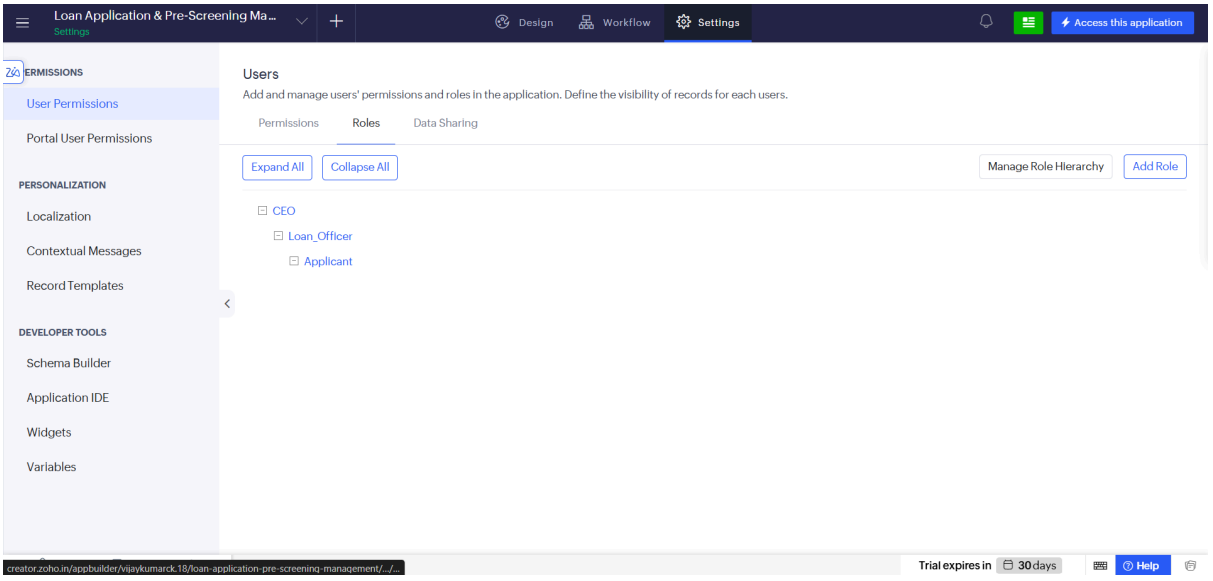
These steps were not completed during this session.

3. Output

The configured RBAC structure is summarized below.

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Component	Configured Setting	Status
Role Hierarchy	CEO → Loan_Officer → Applicant	Completed
Officer_Permission	View/Edit on form, no Delete, dashboard access	Completed
Applicant_Permission	Form access, no dashboard access	Completed
Test Users	2 users with assigned role and permission	Completed
Data Sharing Rule	Applicant records shared to Loan_Officer (Read/Write)	Completed
Live Login Test	Verification of visibility per role	Pending





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Figure 8: Final configured RBAC summary (Roles hierarchy and Data Sharing tab together)

4. Result

A three-level role-based access control model was configured for the Loan Application & Pre-Screening Management application in Zoho Creator. The CEO, Loan_Officer, and Applicant roles were arranged in a hierarchy, each with a permission profile matching its responsibilities. Two test users were assigned to the Loan_Officer and Applicant roles, and a data sharing rule was created to give Loan Officers Read/Write access to records submitted by Applicants. The configuration satisfies the RBAC requirements of the experiment. Live verification of per-role visibility is planned as a follow-up.

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